

Kaggola Reddy Balaji

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Professional Summary

Computer Vision Engineer with hands-on professional experience at IIT Tirupati Navavishkar I-Hub Foundation, specialized in building and deploying real-time video analytics systems using YOLOv8, PyTorch, and OpenCV. Skilled in face recognition, object tracking, and scalable AI deployments on AWS. Proven ability to deliver production-grade solutions with high accuracy and real-world impact.

Skills

- **Languages:** Python
- **Frameworks:** YOLOv8 (Ultralytics), PyTorch, OpenCV,
- **Computer Vision:** Object Detection, Face Recognition, Tracking (ByteTrack), OCR
- **Techniques:** Transfer Learning, CLAHE, Cosine Similarity, Model Optimization
- **Tools & Platforms:** Roboflow, AWS EC2, Git and GitHub

Experience

Computer Vision Engineer | Technology Innovation Hub, IIT Tirupati (March 2025 - Present)

- Designed and developed an automated face recognition attendance system using a YOLOv8 → InsightFace → AdaFace IR-101 pipeline, eliminating proxy attendance by replacing manual roll calls with camera-based identity verification across 72 registered individuals, achieving **97.71% accuracy** on 22,218 evaluated samples.
- Built a real-time **Staff/Customer tracking system** using YOLOv8 + ByteTrack on indoor camera feeds — tracking idle/active time, queue wait times, and breach alerts via a Flask + Socket.IO dashboard deployed on **AWS EC2**
- Developed a vehicle number plate recognition system using YOLOv8 and EasyOCR for real-time detection and OCR from CCTV footage, improving localization accuracy under variable lighting and weather conditions.

Projects

AI-Based Face Recognition Attendance System | YOLOv8 · InsightFace · AdaFace IR-101 · ByteTrack · CLAHE · Flask

- Designed a multi-stage pipeline for camera-based attendance marking across 72 individuals, eliminating proxy attendance in a real-world college environment
- Achieved 97.71% accuracy and 95.10% macro F1 on 22,218 evaluated samples with cosine similarity matching and temporal voting for robust identity verification

GitHub: <https://github.com/balaji2345/face-recognition-adaface>

Vision Based Employee Tracking Under Indoor Environments | YOLOv8 · stableTracker · Flask · Socket.IO · AWS EC2

- Developed a real-time indoor tracking system using YOLOv8, achieving **95.34% mAP@0.5** on a custom dataset
- Developed a custom multi-class tracker (IoU + centroid + re-ID) to handle occlusions and eliminate ghost IDs
- Deployed on AWS EC2 with Flask + Socket.IO for live video, KPI dashboard, and real-time alerts

GitHub: <https://github.com/balaji2345/employee-tracking-in-indoor-environments>

Smart VNPR Access Control System (Final Year Project — Published Research) | Raspberry Pi · Pi Camera · ANPR · OCR · Servo Motors

- Designed and implemented a vehicle number plate recognition system for automated barrier control in high-security buildings, using ANPR algorithms and a centralized authorized plate database.

Education

B.Tech, Electronics and Communication Engineering | Sri Venkateswara College of Engineering (2021 - 2025)

- **CGPA - 8.33**

Certifications

- Python Programming Masterclass (30 Days) — Pantech e Learning, 2023
- TCS ion Career Edge – Young Professional — TCS ion, 2024
- Communication Skills — TCS ion, 2024